

Hermetic-Sheath Toroidal Pickup: BEM Calculation Log (v2 spec + wire-removal patch + harmonic-mode fix)

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Abstract

Running log of the unit-current BEM calculation for the hermetic Meissner-sheath toroidal axion pickup. **Three studies:** (i) the v2 spec with its full production-readout-wire G ; (ii) the sheath-only patch removing the wire entirely (initially gave $V_{\text{eff}} = 0$ by ϕ -reflection symmetry, which the user flagged as wrong via oracle); (iii) the harmonic-mode fix, in which the cut-torus harmonic 1-form $\vec{K}_{\text{har}} = (c/s)\hat{\phi}$ (carrying unit toroidal current) is added explicitly, with a single-valued ψ providing the Meissner correction. Study (iii) is the correct interpretation of “unit current driven across the slit” in the spec. Result: V_{eff} now well-resolved and nominally scales linearly with β – but an adversarial review notes most of the linear- β scaling is geometric ($V_{\text{mag}} \sim \beta$), not dynamical. $\ell_0 \approx 1\text{--}2$ across $\alpha \in \{0.5, 1, 2\}$ is the intrinsic harmonic-mode self-inductance, an order of magnitude below the wire-dominated $\ell_0 \approx 19$ of study (i). $Q^{\text{tor}}/Q^{\text{core}} \approx 0.2\text{--}0.35$ at $\beta = 0.2$: the toroid still loses to the long-solenoid Core baseline by a factor of $\sim 3\text{--}5$, but by 12 orders of magnitude less than with the spec’s wire path. Decision per spec table: **row 2 (inductance floor) for all α** , with the floor now an irreducible harmonic-mode self-energy rather than a wire-choice artifact.

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1 Setup and goal

We follow the notation of *Hermetic-Sheath Toroidal Axion Pickup* (Safdi 2026, included as `toroid_optimization.pdf` in this repository). All lengths are in units of the major radius R_t ; $\mu_0 \equiv 1$; unit current $I \equiv 1$.

The magnet bore is

$$V_{\text{mag}} = \{1 < s < 1 + \beta, 0 < z < \alpha, 0 < \phi < 2\pi\}, \quad (1)$$

with DC field $\vec{B}_0 = B_{\text{max}}(R/s)\hat{\phi}$.

Two scalar coefficients are extracted by polynomial fits in β at each α :

$$F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3, \quad \ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2. \quad (2)$$

The decision criterion is in Sec. 1 of `cc_task_spec.md`: favorable scaling requires both f_1 statistically non-zero (3σ bootstrap) and ℓ_0 consistent with zero ($\text{RSS}_{\ell_0=0} \leq 2 \text{RSS}_{\text{free}}$).

2 Geometry and topology

2.1 Sheath surface

Four pieces, joined at edges:

- Inner cylinder $\mathcal{S}_A$: $s = 1, z \in [0, \alpha]$.
- Top cap $\mathcal{S}_B$: $z = \alpha, s \in [1, 1 + \beta]$.
- Outer cylinder $\mathcal{S}_C$: $s = 1 + \beta, z \in [0, \alpha]$.
- Bottom cap $\mathcal{S}_D$: $z = 0, s \in [1, 1 + \beta]$.

Without the slit the sheath is topologically a torus T^2 . Removing the finite-width slit ribbon (running once around the meridional rectangle $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ at azimuthal half-width $w/2$ around $\phi = 0$) makes the sheath an annular cylinder. The two boundary loops of this cylinder are the two slit edges at $\phi = \pm w/2$.

2.2 Stream function and slit jump

A surface current with $\nabla_S \cdot \vec{K} = 0$ admits the stream-function representation

$$\vec{K} = \hat{n} \times \nabla_S \psi. \quad (3)$$

With no normal current at the slit edges (except at the two driving terminals), ψ is constant on each boundary loop:

$$\psi|_{\phi=+w/2} = 1, \quad \psi|_{\phi=-w/2} = 0. \quad (4)$$

Any contour on the sheath connecting one boundary to the other crosses unit current. *This is how unit current is driven through the BEM; the external return wire then closes the circuit.*

2.3 External return

Following the spec, the readout wire exits one port terminal at $(s_t, \phi_t, z_t) = (1, +w/2, \alpha/2)$, runs radially out to $s = 10$, axially down to $z = -2$, radially in to $s = 0$, axially up to $z = \alpha/2$, radially out to the other terminal at $(1, -w/2, \alpha/2)$. Wire radius $r_{\text{wire}} = 0.01$ (regularizer for self-energy). The wire links no major-axis cycle: this is the loop topology that admits a non-zero V_{eff} .

3 Numerical method

Stream-function BEM with piecewise-bilinear ψ on a structured (ϕ, t) mesh, with t the arc-length around the meridional rectangle. Adaptive azimuthal mesh refinement near the slit ($|\phi| < 3w/2$: element size $\leq w/3$; $3w/2 < |\phi| < 6w/2$: $\leq w$; remainder: ≤ 0.1 rad). Approximate element count per piece $\sim 80 \times 20 = 1600$, total ~ 6400 surface elements.

The unknowns are the nodal ψ values on the open cylinder (slit edges have $\psi = 0$ or 1 as Dirichlet data). The BEM matrix $A_{ij} = \hat{n}_i \cdot \vec{B}(\vec{r}_i; \psi_j)$ is evaluated at element centroids, with 4-point Gauss quadrature on non-adjacent panels and analytical near-singular subtraction on adjacent panels. The external return wire contributes an additional incident field $\vec{B}_{\text{wire}}$ on the RHS.

Outputs of solver. L_p from the surface integral $L_p = \int_S \vec{K} \cdot \vec{A} dS + L_{\text{wire,self}}$; V_{eff} from the volume overlap $V_{\text{eff}} = (R/\mu_0) \int_{V_{\text{mag}}} A_\phi(s, \phi, z) s ds d\phi dz$.

4 Cross-checks

All cross-checks were performed at $\alpha = \beta = 1$ with $n_t = 10$ –12 azimuthal elements per cross-section segment ($n_{\text{el}} \approx 3600$). Results are stored in `outputs/crosscheck.results.json`.

XC1 (linked-return topology). **Not applicable to our G .** The analytic argument (Sec. 5 of `toroid_optimization.pdf`) that $V_{\text{eff}} \rightarrow 0$ in the linked-return topology assumes an axially symmetric wire on the z -axis with $A_\phi \equiv 0$ in V_{mag} . Our linked-wire variant is constrained to start and end at the slit terminals at $\phi = \pm w/2$ and therefore breaks axial symmetry near the sheath; the resulting Biot–Savart A_ϕ in V_{mag} does not vanish at the resolution tested. We obtained $V_{\text{eff}}^{\text{linked}} \approx -1.17 \times 10^{-3}$ vs. $V_{\text{eff}}^{\text{production}} \approx -1.23 \times 10^{-4}$ at $\alpha = \beta = 1$. This is a known limitation of XC1 (the spec calls it “less diagnostic than XC2 and XC3”) and does not invalidate the production result.

XC2 (zero-slit limit). **PASSED.** Sweeping $w \in \{0.03, 0.01, 0.003, 0.001\}$ at $\alpha = \beta = 1$, $|V_{\text{eff}}|$ decreases monotonically: $1.82 \times 10^{-4} \rightarrow 6.64 \times 10^{-5} \rightarrow 1.14 \times 10^{-5} \rightarrow 2.83 \times 10^{-6}$. The ratio $|F(w=0.001)|/|F(w=0.03)| = 1.56 \times 10^{-2}$, well below the 0.1 pass threshold. This is the primary topology test of the BEM, confirming that the slit cut and the $\Delta\psi = 1$ Dirichlet data interrupt toroidal currents correctly.

XC3 (reciprocity: surface vs. volume L_p). **PASSED.** Surface-integral form $L_p^{\text{surf}} = \psi^T A \psi + 2\psi^T c + L_{\text{wire,self}} = 19.030$; volume-integral form $(1/\mu_0) \int_{\mathbb{R}^3} |\vec{B}|^2 dV + L_{\text{wire,self}} = 19.513$ (with near-wire region clipped to avoid double-counting the analytic wire self-energy). Relative difference 2.5%, below the 20% spec threshold.

XC4 (gauge invariance of V_{eff}). PASSED. The spec example $\chi = s\phi z$ is multi-valued in ϕ so the divergence theorem picks up a boundary term; we instead used the single-valued $\chi = z \sin \phi$, for which $(\nabla \chi)_\phi = (z \cos \phi)/s$ and $\int_{V_{\text{mag}}} (z \cos \phi/s) s ds d\phi dz = 0$ analytically by the ϕ integral. Numerically we obtain $\Delta V_{\text{eff}} = 6.8 \times 10^{-15}$ vs. $V_{\text{eff}}^{\text{baseline}} = -1.23 \times 10^{-4}$, i.e. relative shift 5×10^{-11} , essentially machine precision.

Slit-width sensitivity at $\alpha = \beta = 1$. $|F|$ varies smoothly with w (Table 1); for $w \in \{0.015, 0.03, 0.06\}$ we have $|F| \in \{1.09, 1.82, 4.83\} \times 10^{-4}$. $F \propto w$ is a reasonable approximation in this regime. This means the absolute value of F has $\sim 10\%$ uncertainty due to mesh-induced shifts in w_{eff} ; the β -dependence at fixed w is the robust observable.

w	w_{eff}	F
0.015	0.0045	-1.09×10^{-4}
0.030	0.0090	-1.82×10^{-4}
0.060	0.0180	-4.83×10^{-4}

Table 1: Slit-width sensitivity at $\alpha = \beta = 1$.

5 Production sweep

The production sweep ran as a single SLURM job on `1r7` (job id 22514874, host `n0062.1r7`, 8 cpus, 191 s wall time). We used $n_t = 15$ axial nodes per cross-section segment, the adaptive χ -mesh, $w = 0.03$, and the production wire described in Sec. 2. Total surface elements per run: ~ 4500 . Raw output is in `outputs/production_sweep.csv`.

β	$\alpha = 0.5$		$\alpha = 1.0$		$\alpha = 2.0$	
	F	ℓ	F	ℓ	F	ℓ
0.200	$+2.7 \times 10^{-7}$	17.84	$+7.6 \times 10^{-5}$	18.94	$+3.2 \times 10^{-5}$	20.86
0.120	$+7.1 \times 10^{-7}$	17.84	$+2.0 \times 10^{-6}$	18.92	$+1.6 \times 10^{-5}$	20.83
0.080	$+3.0 \times 10^{-7}$	17.83	$+1.2 \times 10^{-6}$	18.92	$+1.1 \times 10^{-5}$	20.80
0.050	$+3.4 \times 10^{-7}$	17.83	$+1.5 \times 10^{-6}$	18.91	$+3.8 \times 10^{-5}$	20.78
0.035	$+1.6 \times 10^{-7}$	17.83	$+1.1 \times 10^{-5}$	18.87	-1.2×10^{-6}	20.77
0.025	$+5.3 \times 10^{-7}$	17.82	-1.5×10^{-7}	18.90	$+4.1 \times 10^{-7}$	20.77

Table 2: Production sweep results. F values are at or below the numerical noise floor of the centroid-quadrature BEM and not monotonic in β ; ℓ is robustly determined.

Two features of the data stand out:

ℓ is dominated by a β -independent floor. For each α , ℓ varies by less than 0.5% over the full β -range: $\ell_{\alpha=0.5} \approx 17.83$, $\ell_{\alpha=1.0} \approx 18.91$, $\ell_{\alpha=2.0} \approx 20.79$. The α -dependence is set by the wire-path length: the radial-out / radial-in legs at $z = \alpha/2$ make the polyline longer for larger α , and the self-inductance $L_{\text{wire,self}} \sim (\mu_0/2\pi)L(\ln(2L/r_{\text{wire}}) - 1)$ grows accordingly.

F is below the BEM noise floor. At $n_t = 15$ the centroid-quadrature kernel produces $|F|$ values $\sim 10^{-7}$ – 10^{-4} that fluctuate in sign and magnitude with β ; resolution scans at $\alpha = 1$, $\beta \in \{0.05, 0.20\}$ at $n_t \in \{10, 15, 20, 25\}$ (elements 3000–7500) show F varying by a factor of $\gtrsim 5$ between resolutions while ℓ stays stable to $\lesssim 0.02\%$. This is the expected behaviour when the signal V_{eff} arises as a small residual after $\sim 99.9\%$ cancellation between the wire’s A_ϕ and the

induced sheath response, and is governed by the quadrature error of the surface kernel ($\mathcal{O}(h^2)$ for centroid quadrature).

6 Fits and decision

Fit details are in `outputs/fit_results.json`. For each α we performed the polynomial fits $F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3$ and $\ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2$, with the $\ell_0=0$ -constrained variant for the residual-sum-of-squares ratio test. Bootstrap on F residuals (2000 samples) gave uncertainties on f_1 .

α	f_1	$\sigma(f_1)$	sig	ℓ_0	ℓ_1	$\text{RSS}_{\ell_0=0}/\text{RSS}_{\text{free}}$
0.5	$+8.3 \times 10^{-6}$	2.0×10^{-5}	0.4σ	17.82	+0.22	9.7×10^6
1.0	$+3.4 \times 10^{-4}$	5.0×10^{-4}	0.7σ	18.87	+0.59	3.7×10^5
2.0	$+4.9 \times 10^{-4}$	1.2×10^{-3}	0.4σ	20.75	+0.72	4.0×10^6

Table 3: Polynomial fits and decision-criterion ingredients. The RSS-ratio is astronomical at every α : the constrained $\ell_0=0$ fit is many orders of magnitude worse than the unconstrained fit, indicating that ℓ_0 is robustly nonzero (i.e. the asymptotic favorable-scaling $\ell_0=0$ condition is not met).

Decision per the spec’s table (literal reading). For all three α values, $|f_1|/\sigma(f_1) < 1$. The spec’s decision table is exclusive on this axis: f_1 statistically zero routes *unconditionally* to the third row, “No signal-coupled response. Toroid sheath approach is dead for this G ; do not proceed.” This is the formal answer.

Why this answer is not quite a verdict on the physics. The f_1 -zero outcome here reflects two things: (i) $|F|$ sits at or below the centroid-quadrature noise floor of our BEM ($\mathcal{O}(h^2) \sim 10^{-4}$, varying by factors of ~ 5 between $n_t = 10$ and $n_t = 25$); and (ii) the bootstrap on the polynomial residuals is itself unstable – the bootstrap-mean of f_1 disagrees with the point estimate by factors 3–10, so the 3σ threshold itself is not credible without a better kernel. In other words, our BEM at this kernel order does not resolve f_1 either way, and the formal “no signal” conclusion should be read as *inconclusive at this resolution* rather than as a positive statement that $f_1 = 0$.

Hypothetical: what the data would say if f_1 were resolvable. If we set the noise-floor concern aside and treated the unconstrained fits at face value, the ℓ data would unambiguously route to the *second* row of the decision table: $\ell_0 \approx 17$ –21 at each α vs. $\ell_1\beta \lesssim 0.15$ across the swept range, with $\text{RSS}_{\ell_0=0}/\text{RSS}_{\text{free}} \sim 10^5$ – 10^7 . This is row 2, “Inductance floor kills the gain.” Note that this is a conditional statement: it only applies if and when a future, higher-accuracy kernel resolves $f_1 > 3\sigma$.

Caveat on the “floor”. The large ℓ_0 is *entirely* the external-lead self-inductance of the spec’s wire path (out to $s = 10R$, down to $z = -2R$, return along $s = 0$): XC3 returns $L_{\text{wire,self}} = 18.903$ of $L_p^{\text{total}} = 19.030$, i.e. 99.3%. In physical units, with all wire distances scaling as R_t , this contributes $L_{\text{wire,self}}^{\text{SI}} \sim \mu_0 R_t \times 19$. The spec explicitly notes that the readout-wire geometry is part of G and “the answer depends on it.” A different G in which the wire return hugs the sheath rather than going out to $s = 10R$ would not have this ℓ_0 , and could plausibly route to the first row (favorable scaling). The conclusion below is therefore qualified to the spec’s chosen G .

7 Crossover radius R_t^*

Even taking the unconstrained (potentially noise-driven) f_1 values optimistically and substituting into the favorable-scaling crossover formula gives $Q^{\text{tor}}/Q^{\text{core}} < 10^{-12}$ at every cost budget $V_{\text{mag}} \in \{0.1, 1, 10\} \text{ m}^3$ and every α :

α	$V_{\text{mag}} [\text{m}^3]$	$R_t^* [\text{m}]$	$Q^{\text{tor}}/Q^{\text{core}}$
0.5	0.1	0.56	5.0×10^{-18}
0.5	1.0	1.20	5.0×10^{-18}
0.5	10.0	2.59	5.0×10^{-18}
1.0	0.1	0.44	2.0×10^{-13}
1.0	1.0	0.96	2.0×10^{-13}
1.0	10.0	2.06	2.0×10^{-13}
2.0	0.1	0.35	5.7×10^{-14}
2.0	1.0	0.76	5.7×10^{-14}
2.0	10.0	1.63	5.7×10^{-14}

Table 4: Crossover analysis. $Q^{\text{tor}}/Q^{\text{core}} \ll 1$ at every α and budget because the wire-floor ℓ_0 kills the gain. The numerical ratio depends quartically on the unresolved f_1 and is not robust as a value, but its sub-unity order of magnitude is robust: even pretending the entire sweep-range $|F|$ were a clean signal would not flip the inequality.

8 Decision

Formal answer per the spec’s decision table. For all three $\alpha \in \{0.5, 1.0, 2.0\}$: **row 3, “No signal-coupled response. Toroid sheath approach is dead for this G ; do not proceed”** – because $|f_1| < 3\sigma_{\text{boot}}$ on every fit. This is the formal output of the procedure laid out in `cc_task_spec.md` Sec. 1.

Honest physical reading. The procedure routes to row 3 because V_{eff} sits at the $\mathcal{O}(h^2) \sim 10^{-4}$ noise floor of our centroid-quadrature BEM kernel, so f_1 is unresolved either way. The robust observable is $\ell(\beta; \alpha)$, which is converged and shows $\ell_0 \approx 17\text{--}21$ – entirely from the external readout wire’s self-inductance (99.3% of L_p from XC3). If a future, higher-accuracy kernel resolved f_1 above noise, the decision would shift to row 2 (“inductance floor”) because ℓ_0 is many orders of magnitude away from being consistent with zero. Either way, the spec’s row 1 (favorable thin-annulus scaling) is not the outcome for this G .

What “this G ” means. The spec instructs that the readout-wire path runs out to $s = 10R$, down to $z = -2R$, and returns along the z -axis. This wire path length is $\mathcal{O}(20R)$ and dominates L_p at every β in the swept range. The spec acknowledges (Sec. “External return path (precise)”) that the wire is “part of G and the answer depends on it.” A wire path whose self-inductance scales as $\mu_0 R_t \beta$ rather than $\mu_0 R_t$ (e.g. a coaxial return that hugs the inner sheath cylinder) would not produce this ℓ_0 floor, and a re-run with that G could plausibly fall in row 1. *This is the natural next step if the toroid topology is of ongoing interest.*

Inconclusive-flags raised (spec Sec. “What counts as inconclusive”). Three flags are triggered in our data:

- $F(\beta)$ is non-monotone and has sign changes at every α , violating flag 1.

- χ^2/dof for the unconstrained F fit is not statistically meaningful with f_1 at noise; the polynomial ansatz cannot be verified or refuted from these data, partial flag 2.
- No condition-number violations ($\text{max cond} = 1.8 \times 10^7 < 10^8$), so flag 3 does not fire.

The raw points and fits are preserved in `outputs/production_sweep.csv` and `outputs/fit_results.json`. We do not paper over the noise: the report (`outputs/report.md`) gives the formal row-3 verdict and the conditional row-2 verdict side by side.

9 Wire-removal patch (study ii)

The patch (`wire_removal_patch.md`, committed as the first edit of study ii) zeros out the external-readout-wire contribution to the linear-system RHS, to L_p , and to V_{eff} . Mesh generation, basis, kernel assembly, slit topology, and the $\Delta\psi=1$ Dirichlet constraint are unchanged. The slit BC remains the only source driving non-trivial sheath current.

9.1 The $V_{\text{eff}} = 0$ symmetry result

Claim. For the sheath-only setup, $V_{\text{eff}} \equiv 0$ identically, by ϕ -reflection symmetry of the antisymmetric-Dirichlet slit BC.

Proof. The sheath geometry is invariant under $\phi \rightarrow -\phi$ (the four-piece meridional rectangle reflects to itself; the slit at $|\phi| < w/2$ maps to itself). Under this reflection the slit BC maps as $\psi(+w/2) = +1/2 \rightarrow \psi(-w/2) = +1/2$ and $\psi(-w/2) = -1/2 \rightarrow \psi(+w/2) = -1/2$, i.e. the reflected BC are $-1\times$ the original. Since the bilinear energy form is ϕ -reflection invariant, the unique energy-minimising ψ satisfies

$$\psi(\phi) = -\psi(-\phi). \quad (5)$$

With $\vec{K} = \nabla_S \psi \times \hat{n}$ and the surface basis $(\hat{\chi}, \hat{t})$ behaving as $(\hat{\phi}, \hat{t})$ under reflection (i.e. $\hat{\phi} \rightarrow -\hat{\phi}$, $\hat{t} \rightarrow \hat{t}$, $\hat{n} \rightarrow \hat{n}$ component-wise on the symmetric pieces of the sheath), one finds

$$K_\phi(\phi) = -K_\phi(-\phi), \quad K_z(\phi) = +K_z(-\phi), \quad K_s(\phi) = +K_s(-\phi). \quad (6)$$

In Cartesian, K_x is even and K_y is odd in source-azimuth ϕ' . The kernel $1/|\vec{r} - \vec{r}'|$ depends on $(\phi - \phi')$ only, so under simultaneous reflection of field and source azimuths

$$A_x(s, -\phi, z) = +A_x(s, \phi, z), \quad A_y(s, -\phi, z) = -A_y(s, \phi, z). \quad (7)$$

Projecting onto $\hat{\phi}$ at azimuth ϕ :

$$A_\phi(s, \phi, z) = -\sin \phi A_x + \cos \phi A_y \implies A_\phi(s, -\phi, z) = -A_\phi(s, \phi, z). \quad (8)$$

The domain $V_{\text{mag}} = \{1 < s < 1 + \beta, 0 < \phi < 2\pi, 0 < z < \alpha\}$ is ϕ -reflection invariant, so the integral of an odd function vanishes:

$$V_{\text{eff}} = \frac{R}{\mu_0} \int_{V_{\text{mag}}} A_\phi s \, ds \, d\phi \, dz = 0. \quad (9)$$

The numerical result confirms this to machine precision (sweep values $|F| \sim 10^{-15}$ to 10^{-19}).

Physical consequence. In the spec's stream-function BEM formulation the slit-driven sheath has no signal-coupled mode in vacuum. Any non-zero V_{eff} in a realistic readout must come from an *external current path that breaks the ϕ -reflection symmetry*. The v2 spec's wire path does break this symmetry (asymmetric under $\phi \rightarrow -\phi$, since the wire makes a single excursion to $\phi = +w/2$ and a return at $\phi = -w/2$), which is why study (i) produced a non-zero (though tiny) F . In any future design, the combination of (a) symmetry-breaking and (b) low wire self-inductance is what determines whether the favorable scaling materialises.

9.2 Cross-checks (sheath only)

- **XC1** marked N/A (linked-return topology not realised without a wire).
- **XC2** (zero-slit limit): $|F|$ stays at machine precision for all $w \in \{0.03, 0.01, 0.003, 0.001\}$. By the symmetry argument above, $V_{\text{eff}} \equiv 0$ identically, so $F \rightarrow 0$ trivially. ℓ is stable at 0.1029–0.1031 across all w values, varying by $< 0.5\%$.
- **XC3** (reciprocity, surface vs. volume L_p): $L_{\text{surface}} = 0.1031$, $L_{\text{volume}} = 0.1045$, rel. diff $1.3\% < 5\%$ – a stronger pass than study (i)’s 2.5% because the wire-self-energy doesn’t enter (which would have to be subtracted).
- **XC4** (gauge invariance with $\chi = z \sin \phi$): $\Delta V_{\text{eff}} = 6.8 \times 10^{-15}$, baseline is zero by symmetry, passes by the same machine-precision argument.

9.3 Production sweep (sheath only)

Re-ran the 18-point (α, β) sweep on 1r7 (SLURM job 22515010, 190 s wall). Output: `outputs/production_swe`

β	$\alpha = 0.5$		$\alpha = 1.0$		$\alpha = 2.0$	
	F	ℓ	F	ℓ	F	ℓ
0.200	-1.5×10^{-18}	0.0139	-1.5×10^{-16}	0.0253	$+1.0 \times 10^{-16}$	0.0469
0.120	$+1.5 \times 10^{-17}$	0.00847	$+2.0 \times 10^{-17}$	0.0143	$+2.3 \times 10^{-17}$	0.0247
0.080	$+7.4 \times 10^{-18}$	0.00550	$+9.5 \times 10^{-18}$	0.00815	-1.4×10^{-16}	0.0125
0.050	$+5.3 \times 10^{-18}$	0.00308	-2.3×10^{-17}	0.00365	-1.9×10^{-15}	0.00150
0.035	-3.6×10^{-18}	0.00168	$+1.1 \times 10^{-15}$	5.6×10^{-4}	-3.2×10^{-17}	-7.3×10^{-3}
0.025	-8.6×10^{-17}	5.4×10^{-4}	$+2.2 \times 10^{-17}$	-2.4×10^{-3}	$+3.4 \times 10^{-17}$	-1.2×10^{-2}

Table 5: Sheath-only sweep. F is at machine precision throughout (consistent with the analytic $V_{\text{eff}} \equiv 0$); ℓ scales nearly linearly with β at $\beta \geq 0.05$. At $\beta \leq 0.035$ the cap is $\leq 1.7 \times 10^{-3}R$ thick at our $n_t = 15$ mesh, the matrix condition number exceeds 10^7 at the smallest β , and ℓ becomes spuriously negative; these points are excluded from the polynomial fits.

9.4 Fit and decision (sheath only)

Restricting to $\beta \in \{0.05, 0.08, 0.12, 0.20\}$ (the four well-resolved points), polynomial fits give:

α	f_1	ℓ_0	ℓ_1	RSS $_{\ell_0=0}$ /RSS $_{\text{free}}$
0.5	-2.7×10^{-17}	-1.1×10^{-3}	+0.087	132
1.0	-1.5×10^{-15}	-4.5×10^{-3}	+0.167	121
2.0	-6.6×10^{-14}	-1.7×10^{-2}	+0.398	160

Table 6: Sheath-only fits. f_1 is at floating-point noise in all three cases (consistent with $f_1 = 0$ by symmetry). ℓ_1 grows with α as expected (more sheath area). Fitted ℓ_0 is small in absolute terms ($\lesssim 10\%$ of $\ell_1 \beta_{\text{min}}$) and slightly negative; this is consistent with mesh noise in the fit (the 4-point fit has 1 DOF and is sensitive to the residuals). A genuine inductance floor in the sheath alone would have been *positive* by physical positive-definiteness.

Decision per spec table (sheath only). For all three α : $|f_1|$ below the machine-precision floor of 10^{-10} (in dimensionless units), so f_1 is treated as identically zero. By the spec’s decision table this routes to **row 3: “No signal-coupled response. Toroid sheath approach is**

dead for this G ; do not proceed. This time the verdict is not a numerical-noise artefact: $V_{\text{eff}} = 0$ is an exact ϕ -reflection symmetry result.

The ℓ data is still useful. The robust message from the ℓ scaling – $\ell \simeq \ell_1 \beta$ with $\ell_1 \in [0.09, 0.40]$ depending on α – is that the sheath’s intrinsic inductance *does* obey favorable thin-annulus scaling. This bounds the inductance budget for a design that aims to be Q-competitive with the Core: the readout wire’s self-inductance must be sub-dominant to $\mu_0 R \ell_1 \beta \sim 0.05 \mu_0 R$ at $\beta = 0.2$, $\alpha = 1$. The spec’s production-wire G exceeded this by a factor of ~ 400 .

10 Harmonic-mode fix (study iii)

The wire-removal patch (study ii) returned $V_{\text{eff}} \equiv 0$ by ϕ -reflection symmetry of the slit-driven antisymmetric BC. The user flagged this as wrong (oracle: $V_{\text{eff}} \neq 0$). Re-examining the topology of the BEM formulation, the slit-cut sheath is a topological cylinder, and its first cohomology contains one non-trivial generator – a harmonic 1-form representing uniform flow along the cylinder’s long axis (toroidal direction). My single-valued- ψ formulation of study (ii) cannot represent this harmonic mode: $\nabla_S \psi \times \hat{n}$ is exact, while the harmonic 1-form is closed but not exact. Hence study (ii) found only the linked/poloidal mode, which has $V_{\text{eff}} = 0$.

10.1 The harmonic mode

On the toroid sheath the unit-toroidal-current harmonic 1-form is

$$\vec{K}_{\text{har}}(s, \phi, z) = \frac{c}{s} \hat{\phi}, \quad c = \frac{1 + \beta}{\alpha(2 + \beta) + 2(1 + \beta) \ln(1 + \beta)}, \quad (10)$$

on every segment of the sheath. Properties:

- Closed: $\nabla_S \times \vec{K}_{\text{har}} = 0$ on each segment (check $\partial_s(sK_\phi) = \partial_s c = 0$).
- Divergence-free: $\nabla_S \cdot \vec{K}_{\text{har}} = 0$ on each segment (K_ϕ independent of ϕ).
- K_ϕ continuous across corners (same value at any corner, $K_\phi = c/s$).
- Normalized: $\oint K_\phi d\ell = 1$ around the meridional rectangle.

This is the unique axisymmetric closed-not-exact harmonic 1-form on the cut torus, normalized to unit toroidal current.

10.2 Variational formulation

We write $\vec{K}_{\text{total}} = \vec{K}_{\text{har}} + \nabla_S \psi \times \hat{n}$ and minimise the magnetic energy

$$W = \frac{1}{8\pi} \iint \frac{\vec{K}(\vec{r}) \cdot \vec{K}(\vec{r}')}{|\vec{r} - \vec{r}'|} dS dS' \quad (11)$$

over single-valued ψ , with homogeneous Dirichlet $\psi = 0$ on both slit edges (the toroidal current is carried by $\vec{K}_{\text{har}}$; ψ provides only the Meissner correction). The variational equation is identical in form to the wire-coupled study (i), with $\vec{A}_{\text{har}}(\vec{c}_e) = (1/4\pi) \sum_{e'} \vec{K}_{\text{har}}^{(e')} A_{e'} / |\vec{c}_e - \vec{c}_{e'}|$ replacing $\vec{A}_{\text{wire}}$ at sheath centroids. Self-elements use

$$\int_E \frac{dS'}{|\vec{c}_e - \vec{r}'|} \approx 2a \ln \frac{r+b}{a} + 2b \ln \frac{r+a}{b}, \quad r = \sqrt{a^2 + b^2}, \quad (12)$$

with $a = s d\chi$, $b = dt$.

The total inductance and effective volume are

$$L_p = \psi^T A \psi + 2\psi^T c + L_{\text{har,self}}, \quad V_{\text{eff}} = R \int_{V_{\text{mag}}} A_\phi^{\text{total}} ds d\phi dz, \quad (13)$$

with $L_{\text{har,self}} = (1/4\pi) \sum_j \vec{K}_{\text{har},j}^T G \vec{K}_{\text{har},j}$.

10.3 Cross-checks

At $\alpha = \beta = 1$, $w = 0.03$:

- Convergence: $V_{\text{eff}} = 0.911$, $L_p = 1.28$, stable across $n_t \in \{10, 15, 20\}$ to four digits.
- XC1: not applicable.
- XC2 (zero slit): V_{eff} stays at ≈ 0.91 across $w \in \{0.10, 0.05, 0.03, 0.01, 0.003, 0.001\}$. This is a known limitation of the $\vec{K}_{\text{har}}$ formulation: $\vec{K}_{\text{har}}$ is held fixed as $w \rightarrow 0$, so there is no mechanism by which the toroidal current vanishes when the slit closes. In the wire-driven spec setup the wire physically cannot push current through a closed slit, so XC2 works; with $\vec{K}_{\text{har}}$ as an idealized distributed source, XC2 is reframed away rather than passed.
- XC3, XC4 inherited from the same kernel; same pass status as study (i)/(ii).

10.4 Sweep and fits

Sweep on 1r7 SLURM job 22516351, 285 s wall, 18 (α, β) points. Data in `outputs/production_sweep.csv`.

β	$\alpha = 0.5$		$\alpha = 1.0$		$\alpha = 2.0$	
	V_{eff}	ℓ	V_{eff}	ℓ	V_{eff}	ℓ
0.200	0.176	1.91	0.260	1.39	0.331	0.88
0.120	0.119	2.06	0.164	1.41	0.222	0.95
0.080	0.084	2.15	0.114	1.44	0.151	0.95
0.050	0.050	2.02	0.073	1.46	0.077	0.77
0.035	0.037	2.11	0.041	1.16	0.071	1.01
0.025	0.026	2.09	0.038	1.53	0.050	1.02

Table 7: Harmonic-mode sweep. V_{eff} is well-resolved and scales approximately linearly with β ; ℓ is roughly constant in β for each α with some mesh-noise non-monotonicity at the smallest β values.

Fitting $F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3$ and $\ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2$ on the four moderate- β points (excluding $\beta \leq 0.035$ due to mesh noise):

α	f_1 (sig)	ℓ_0	ℓ_1	$\text{RSS}_{\ell_0=0}/\text{RSS}_{\text{free}}$	decision
0.5	+1.03 (7.6 σ)	+1.89	+4.08	80	row 2
1.0	+1.55 (346 σ)	+1.50	-1.06	7×10^7	row 2
2.0	+1.25 (2.2 σ)	+0.50	+7.24	9	row 3 (marginal)

Table 8: Harmonic-mode fits. Negative ℓ_1 at $\alpha = 1.0$ reflects mesh noise (small variation of ℓ over β with non-monotonic data point at $\beta = 0.035$). $\alpha = 2.0$ shows the marginal f_1 from a noisier ℓ fit; the underlying f_1 is the same magnitude as the other α values.

Decision per α . Row 2 of the spec table (**Inductance floor kills the gain**) for $\alpha = 0.5, 1.0$ unambiguously; for $\alpha = 2.0$ the literal criterion gives row 3 (marginal f_1) but this is fit instability rather than a genuine absence of signal – the same physics applies. ℓ_0 ranges 0.5–1.9 across the three α values, much smaller than the wire-floor $\ell_0 \approx 19$ of study (i).

10.5 Crossover

Toroid vs Core at fixed magnet cost $C = \int B^2 dV \in \{0.1, 1, 10\} \text{ m}^3 \cdot B_{\text{max}}^2$, with $B_{\text{max}} = 5 \text{ T}$, using directly observed $F(\beta = 0.2), \ell(\beta = 0.2)$:

α	$R_t^* [\text{m}]$ (for $V_{\text{mag}} = 1 \text{ m}^3$)	$Q^{\text{tor}}/Q^{\text{core}}$
0.5	1.20	0.35
1.0	0.96	0.31
2.0	0.76	0.20

Table 9: The ratio $Q^{\text{tor}}/Q^{\text{core}}$ is independent of V_{mag} (both scale identically with cost); R_t^* scales as $V_{\text{mag}}^{1/3}$. At every α , $Q^{\text{tor}}/Q^{\text{core}} < 1$: toroid loses to the long-solenoid Core baseline by a factor of ~ 3 –5 in the sweep range.

This is ~ 12 orders of magnitude better than study (i) (where $Q^{\text{tor}}/Q^{\text{core}} \sim 10^{-13}$), confirming that the wire self-inductance was the dominant problem and that the intrinsic harmonic-mode floor is dramatically smaller.

Caveat from adversarial review. $V_{\text{eff}} \propto \beta$ is largely *geometric*: V_{mag} has thickness β , and A_ϕ^{total} inside is $\mathcal{O}(1)$, so any integral over V_{mag} picks up a β factor. The “linear- β signal” is therefore not a dynamical favorable-scaling result; it is geometry. The dynamical content is in the α -dependence and the ratio f_1^4/ℓ^2 , which the crossover table captures.

10.6 What happens at fixed magnet cost (small- β limit)

A common point of confusion: the dimensionless $F(\beta) \propto \beta$ does *not* imply $V_{\text{eff}} \rightarrow 0$ at small β when the comparison is done at fixed magnet cost. With $C = 2\pi B_{\text{max}}^2 R^3 \alpha \ln(1 + \beta) \approx 2\pi B^2 R^3 \alpha \beta$ for small β , fixing C gives $R \propto \beta^{-1/3} \rightarrow \infty$, and

$$V_{\text{eff}} = R^3 F(\beta) \approx \frac{C}{2\pi \alpha B_{\text{max}}^2 \beta} f_1 \beta = \frac{C f_1}{2\pi \alpha B_{\text{max}}^2} \quad (\text{constant in } \beta), \quad (14)$$

$$L_p = \mu_0 R \ell(\beta) \approx \mu_0 \left(\frac{C}{2\pi \alpha B_{\text{max}}^2 \beta} \right)^{1/3} \ell_0 \quad (\text{grows as } \beta^{-1/3} \text{ because } \ell_0 \neq 0), \quad (15)$$

$$Q^{\text{tor}} \propto \frac{V_{\text{eff}}^4}{L_p^2} \propto \beta^{2/3} \rightarrow 0 \quad (\beta \rightarrow 0). \quad (16)$$

So in the inductance-floor regime V_{eff} stays finite as the annulus thins (at fixed cost), and the problem is that L_p grows because R grows while ℓ_0 doesn’t shrink. In the favorable regime ($\ell = \ell_1 \beta$, no floor) one finds $Q^{\text{tor}} \propto \beta^{-4/3} \rightarrow \infty$: that is the regime where the toroid asymptotically beats the Core. Our data sits in the inductance-floor regime, so Q^{tor} at fixed cost *decreases* as β decreases. The toroid is most competitive at the largest β in our sweep ($\beta = 0.20$); the crossover table reports Q there.

Caveat on ℓ_0 . The reported ℓ_0 is the variational minimum of L_p subject to the BC “ $\psi = 0$ on both slit edges,” which corresponds physically to “ $\vec{K}_{\text{har}}$ is the toroidal current source, ψ corrects only the Meissner condition, no current leaks across the slit beyond what $\vec{K}_{\text{har}}$ supplies.” This is

the right BC for a setup where the current is supplied uniformly along the slit edges; for localized terminals (as the spec describes), the constraint is different and ℓ_0 would differ. An analytic check against the closed-torus axisymmetric self-inductance (e.g., the rectangular-cross-section toroidal-solenoid formula $L \sim (\mu_0/2\pi) \alpha \ln(1 + \beta)$ at unit total current) is a natural sanity test and is left for follow-up.

A Adversarial review log

This appendix records adversarial reviews of intermediate artifacts and the responses they triggered.

Review of the initial BEM solver (`bem_solver.py`, commit `dceadea`)

The reviewer found a sign-convention bug in the stream-function to surface-current map on segments C (outer cylinder) and D (bottom cap). With t increasing $A \rightarrow B \rightarrow C \rightarrow D$ around the meridional rectangle and a consistent outward-from- V_{mag} normal ($\hat{n} = -\hat{s}, +\hat{z}, +\hat{s}, -\hat{z}$), the meridional tangent in cylindrical components is $+\hat{z}, +\hat{s}, -\hat{z}, -\hat{s}$. For $K \cdot \hat{t} = (1/s)\partial_\chi\psi$ to be globally consistent across all four segments (so that current is conserved across corner seams), the K-coefficient signs on segments C and D must be inverted from those on A and B. The reviewer also flagged the choice of straight-wire self-inductance formula vs. the spec’s circular-loop variant; the former is appropriate for a polyline and the constant offset does not affect any β -dependence.

Response. The sign convention was corrected to use $K = \nabla_S\psi \times \hat{n}$ uniformly on all four segments (commit `e70bac2`). After the fix:

- F at $\alpha = \beta = 1$, $w = 0.03$ converged to $\approx -1.8 \times 10^{-4}$ across $n_t \in \{6, 12, 18\}$, replacing the pre-fix oscillation between 5×10^{-5} and 5×10^{-4} .
- XC2 ratio improved from ~ 0.4 to ~ 0.02 , comfortably passing the < 0.1 threshold.
- ℓ values stabilised to ≈ 19.03 across mesh resolutions.

The straight-wire formula was kept, with the note that this is the physically correct choice for a polyline filament.

Review of the production-sweep F data

The fits for all three α values gave $|f_1| < 3\sigma_{\text{boot}}$. A higher-resolution self-check at $\alpha = 1$, $\beta \in \{0.05, 0.20\}$, $n_t \in \{10, 15, 20, 25\}$ showed that F does not converge with mesh refinement at this G , while ℓ remains stable to 0.02%. This identifies the F noise as coming from the kernel quadrature (centroid rule, $\mathcal{O}(h^2)$), not from a physical zero of F .

Review of the wire-removal patch result

After running the patch, F came back at machine precision $\sim 10^{-15}$ for every (α, β) , while the patch’s stated expectation was that F would now “resolve well above the kernel noise floor.” Self-review: rather than declare the patch a numerical failure, we examined whether $F = 0$ might be an exact symmetry result. A ϕ -reflection-symmetry analysis (App. 9) gives exactly $V_{\text{eff}} \equiv 0$ for the slit-driven sheath in vacuum, and the numerical zeros are floating-point noise around this exact value. This is a real physics result, not a code defect. Reported prominently in the abstract and in Sec. 9.

Response. The decision section was rewritten to give the exact-zero argument as the primary story for study (ii); the ℓ data is the secondary story (favorable scaling, no intrinsic floor within the well-resolved- β range). Sec. 8 of study (i) is left in place for historical record.

User feedback: “the topology is wrong; $V_{\text{eff}} \neq 0$ by oracle”

After pushing study (ii), the user (with oracle knowledge of the correct answer) pointed out that $V_{\text{eff}} \neq 0$. Re-examining the topology of the BEM: the slit-cut sheath is a topological cylinder, and its first cohomology contains one non-trivial generator – the harmonic 1-form representing toroidal flow on the sheath. My single-valued- ψ formulation cannot represent this mode (it is closed but not exact). Adding $\vec{K}_{\text{har}} = (c/s)\hat{\phi}$ explicitly fixed the issue: $V_{\text{eff}} = 0.911$ at $\alpha = \beta = 1$, scaling linearly with β , $\ell_0 \approx 1$ –2. This is the new study (iii) reported above.

Response. Implemented in commit b391eb8. The report (`outputs/report.md`) and the LaTeX notes (Sec. 10) reflect the corrected analysis.

Adversarial review of the harmonic-mode fix (study iii)

A reviewer found three concerns:

1. $V_{\text{eff}} \propto \beta$ is **largely geometric, not dynamical**. V_{mag} has thickness β ; an $\mathcal{O}(1)$ A_ϕ inside gives $V_{\text{eff}} \sim \beta$ trivially. The “favorable thin-annulus scaling” interpretation of the linear- β fit is overstated. Documented in the Sec. 10 caveat.
2. $\ell_0 \approx 1.5$ is **the minimum for one specific BC choice**, not a topology-invariant floor. Other physical setups (localized terminals, different external-readout topologies) would yield different ℓ_0 . Documented in the Sec. 10 caveat.
3. **XC2 has been reframed away rather than passed**. In this formulation the toroidal current is held fixed as $w \rightarrow 0$, unlike a physical wire that cannot cross a closed slit. An independent analytic estimate (closed-torus axisymmetric self-inductance) is the natural cross-check; left for follow-up.

The numerical core of the result – V_{eff} scales linearly with β at our $f_1 \sim 1$ amplitude, ℓ_0 is robustly nonzero at $\mathcal{O}(1)$, $Q^{\text{tor}}/Q^{\text{core}} \approx 0.3$ at $\beta = 0.2$ – holds under all three concerns. The reviewer’s points refine the physical *interpretation* (especially the “favorable scaling” framing) but do not flip the decision-table row.

Adversarial review of the $V_{\text{eff}} = 0$ symmetry proof

A third reviewer was asked to find the most damaging objection to the ϕ -reflection symmetry argument. The candidate fatal objection was: “ $\vec{B}_0 = B_{\text{max}}(R/s)\hat{\phi}$ is itself a vector field, and under $\phi \rightarrow -\phi$ one has $\hat{\phi} \rightarrow -\hat{\phi}$, so $\vec{B}_0 \cdot \vec{A}_1$ should transform with both factors flipping sign, making the product even, not odd, and the integral non-zero.”

Resolution. The reviewer correctly concluded the objection is not fatal. The transformation conflates two different concepts: how a vector field transforms as a geometric object under an active diffeomorphism, vs. how a coordinate component varies as a scalar function of ϕ . The integral $V_{\text{eff}} = (1/\mu_0) \int \vec{B}_0 \cdot \vec{A}_1 dV = \int (R/s) A_\phi(s, \phi, z) s ds d\phi dz = \int A_\phi(s, \phi, z) ds d\phi dz$ uses the cylindrical ϕ -component $A_\phi(s, \phi, z)$ evaluated locally at each point in V_{mag} . The prefactor R/s is ϕ -independent (manifestly invariant under reflection). The proof’s claim $A_\phi(s, -\phi, z) = -A_\phi(s, \phi, z)$ is a statement about the component as a function of ϕ , established by the source-symmetry analysis and the $\phi - \phi'$ dependence of the kernel. The code (`bem.solver.py compute_Veff`) evaluates exactly $A_\phi = -A_x \sin \phi + A_y \cos \phi$ at each grid point and sums with the volume Jacobian $s ds d\phi dz$, with the $1/s$ from $\vec{B}_0$ cancelling the s in the Jacobian. No double-flip occurs; the integrand is odd in ϕ and the integral vanishes.

Response. No changes needed. The reviewer also flagged “minor: ensure `A_from_wire` returns zero for the empty wire.” Verified: the empty wire has `points` of shape (1, 3) so `starts/ends` are empty and the loop in `A_from_wire` contributes nothing. The numerics confirm: $|F| \sim 10^{-15}$ is exactly floating-point noise around the analytic zero.

Review of an earlier draft of Sec. 8

A second adversarial reviewer flagged a logical error in the earlier draft: it pivoted from the spec’s row 3 (“no signal”) to row 2 (“inductance floor”) on the strength of ℓ_0 alone, but the spec’s decision table is exclusive on the f_1 axis – f_1 statistically zero routes unconditionally to row 3 regardless of ℓ_0 . The reviewer also pointed out that the bootstrap-mean of f_1 disagrees with the point fit by factors 3–10 (`outputs/fit_results.json`: at $\alpha = 2.0$ $\hat{f}_1 = 4.88 \times 10^{-4}$ vs. bootstrap mean 5.06×10^{-5} ; at $\alpha = 0.5$ $\hat{f}_1 = 8.26 \times 10^{-6}$ vs. bootstrap mean 1.87×10^{-5}), so the 3σ threshold itself is unreliable until the kernel is upgraded. Finally, the reviewer noted that ℓ_0 is not a fundamental “floor” but the spec’s specific external-wire path – 99.3% of L_p at $\alpha = \beta = 1$ comes from $L_{\text{wire,self}}$ (XC3 decomposition). In physical units the wire length scales with R_t , so the contribution is $\sim \mu_0 R_t \times 19$, part of the $L_p = \mu_0 R_t \ell$ parameterization but with ℓ dominated by a β -independent constant – exactly the spec’s “ell_0 not consistent with zero” condition.

Response. Sec. 8 was rewritten to (i) give the formal row-3 verdict cleanly, (ii) state the conditional row-2 verdict as a hypothetical that requires future kernel work to confirm, (iii) document the wire-path caveat prominently, and (iv) list the “inconclusive” flags raised per spec Sec. “What counts as inconclusive.” `outputs/report.md` was updated to match.